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Groundwater Exploration and Management using Geophysical and Remote Sensing Techniques: a Review

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ABSTRACT

The complex and heterogeneous nature of the subsurface poses'challenges in the accurate identification and characterization of the reservoir or groundwater resources. Because of these challenges, exploration methods have changed and evolved throughout history. Integrating remote sensing and geophysical techniques has also emerged as one of the improved approaches for groundwater exploration and management, enabling accurate identification of potential groundwater zones. This paper combines current knowledge of various geophysical techniques and remote sensing approaches and shows how their integration minimizes exploration uncertainties and reduces costs. This review examines several case studies conducted in different geological environments, such as non-sedimentary and sedimentary rocks. It also discusses the difficulties and limitations associated with combining remote sensing and geophysical exploration techniques and the potential for the future. The study results show how integrating geophysical and remote sensing methods can sustainably improve groundwater resource management.

Keywords: *Groundwater Exploration, Remote Sensing, Geophysical Techniques, Groundwater Management, Hydrogeology.*

1.0 INTRODUCTION

Groundwater, an important resource for human consumption, agriculture, and industrial use, is water located beneath the Earth's surface in pore spaces, or fissures, in soil and rock. Its importance, accounting for approximately 95% of freshwater sources in most areas and driving agricultural and industrial development, cannot be overstated (Obimba*et al.*, 2017). However, the exploration and management of groundwater resources faces significant challenges, particularly in regions associated with complex bedrock (Obimba*et al.*, 2017).The rapid increase in the world's population, coupled with urbanization and economic development, has increased groundwater demand (Idowu andOjo, 2024). This growing pressure on groundwater resources, compounded by the impacts of climate change on water availability, highlights the urgent need for efficient exploration and management strategies (Taylor *et al.*, 2013). Understanding groundwater storage dynamics is critical to ensuring the availability, quality and sustainability of this valuable resources.(Groundwater, 2021).

To effectively use and manage groundwater resources, a comprehensive assessment is essential. This includes studying how water occurs in an area, delineating groundwater potential zones and developing sustainable use strategies (MacDonald *et al.*, 2002; Silwal, 2017). However, traditional groundwater exploration methods, such as well drilling and borehole surveys, are often found to be time-consuming and limited in providing a holistic understanding, particularly in regions with inadequate observation capabilities (Ahamed *et al.*, 2022; Ibrahim *et al.*, 2024). This highlights the need for alternative, more efficient approaches.Geophysical and remote sensing techniques have emerged as powerful groundwater exploration and management tools in response to these challenges. These non-invasive approaches offer cost-effective alternatives to traditional methods, enable large-scale surveys, and provide valuable insights into subsurface hydrogeological conditions (Binley *et al.*, 2015). Geophysical methods, including electrical resistivity, electromagnetic, seismic, gravity, and magnetic techniques, have been widely used in groundwater studies for several decades (Nobes, 1996). These methods enable detailed imaging of subsurface structures, facilitate the identification of water-bearing formations and the assessment of groundwater quality and quantity.At the same time, advances in remote sensing technologies such as satellite imagery and aerial photography enable comprehensive spatial coverage and temporal monitoring capabilities that are invaluable for large-scale groundwater studies. Integrating geophysical and remote sensing techniques provides a comprehensive approach to groundwater exploration and management, including hydrogeomorphological analysis, lineament analysis, and detailed groundwater potential maps (S.S. et al., 2012).

This synergistic approach enables the multiscale characterization of aquifer systems, from regional exploration to site-specific investigations, facilitating the development of more accurate hydrogeological conceptual models (Francés *et al.*, 2015). However, challenges remain when applying these techniques, including data interpretation, resolution limitations, and the need for ground truthing and calibration (Brunner *et al.*, 2007). Furthermore, integrating diverse data sets from multiple sources requires interdisciplinary expertise and can be computationally intensive (Sander, 2007).The complexity of the subsurface environment poses further challenges in accurately identifying and characterizing groundwater resources. This complexity arises from factors such as limited data availability, patterns of groundwater use involving numerous individual users, changing user perceptions and behaviors towards sustainable resource allocation and the delayed and widespread consequences of damage to the resource base (Nag, 2008; Burrows, 2018).

Furthermore, due to climate change and its impacts on the water cycle, it is becoming increasingly difficult to maintain groundwater supplies for the foreseeable future (Osumeje*et al.*, 2023). This highlights the critical need for sustainable development and management of groundwater resources, which requires a thorough assessment of groundwater availability, existing use and balance of resources for future use (Silwal, 2017).Given these challenges and the growing importance of groundwater resources, this review aims to summarize current knowledge on applying geophysical and remote sensing techniques in groundwater exploration and management. By examining the principles, advantages and limitations of various methodologies, as well as recent advances in data integration and interpretation, this article aims to provide researchers, practitioners and policy makers with valuable insights into state-of-the-art approaches to sustainable groundwater resource assessment and management.

This paper summarizes current findings on applying geophysical and remote sensing techniques in groundwater exploration and management. It examines the principles, advantages, limitations, and recent advances in integrating and interpreting data from these techniques.

2.0 METHODOLOGY

Literature Review

To properly review groundwater exploration and management using geophysical and remote sensing techniques, an analysis was conducted using Google Scholar, Web of Science and SCOPUS using related terms such as "remote sensing and geophysical techniques" and "remote sensing and groundwater". Groundwater Management", "Geophysical techniques and their applications in groundwater studies", "Remote sensing and their applications in groundwater studies".

Identification, Screening, and Inclusion Criteria

The following identification, screening, eligibility and inclusion criteria will be considered in identifying and selecting the specific literature included in this study. Only titles directly related to groundwater exploration and management using geophysical methods and remote sensing were selected for this study. Only publications from reputable journals published within the last decade were used for the data and information in this study.

Data Synthesis

The data from the above procedures were synthesized to identify the common ideas and the different results in the selected studies.

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Following this methodology, as shown in Fig. 1 below, this review paper aims to provide a comprehensive overview of the current state of knowledge on applying geophysical and remote sensing techniques in groundwater exploration and management.

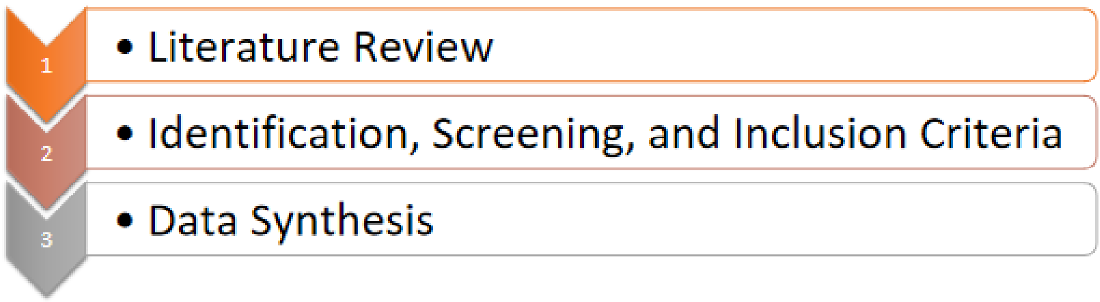

Fig. 1: Methodology flowchart

3.0 OVERVIEW OF GEOPHYSICAL METHODS FOR GROUNDWATER EXPLORATION

3.1 Electrical Resistivity Methods:

Due to their sensitivity to salinity, geophysical methods such as Vertical Electrical Sounding (VES) and Electrical Resistivity Tomography (ERT) are often used in groundwater exploration. They provide valuable information on aquifer geometry and changes in water quality. ERT has significantly advanced data acquisition systems and inversion algorithms, improving its resolution and reliability for aquifer characterization (Loke *et al.*, 2018). Recent applications include high-resolution imaging of coastal aquifers for saltwater intrusion studies (Costall *et al.*, 2020), characterization of karst aquifers and detection of preferred flow paths (Bermejo *et al.*, 2018), and time-lapse monitoring of controlled aquifer recharge processes (Sendrós*et al.*, 2020).

Induced Polarization (IP):

IP is used to investigate aquifer characteristics and estimate hydraulic conductivity (Revil *et al.*, 2015). It can also map contaminated plumes in groundwater and distinguish between clay content and water saturation in aquifers (Nordsiek *et al.*, 2016; Flores-Orozco *et al.*, 2019).

Electromagnetic Methods:

Airborne Electromagnetic (AEM) Surveys:

Auken *et al.*, (2017) state that AEM technology has improved groundwater exploration at a regional scale and provides high-resolution 3D models of subsurface conductivity. The AEM technique exploits natural fluctuations in subsurface electrical conductivity, resulting in fluctuations in rock and pore fluid properties. The conductivity models are integrated with other hydrogeological datasets to conclude groundwater and aquifer properties (Geoscience Australia, 2011).

Ground-based Time-Domain EM (TDEM):

They are based on electromagnetic waves' electrical current induction in the subsurface.

Seismic Methods:

Seismic methods use acoustic waves to image subsurface structures and provide information about the geometry and properties of the aquifer. This is due to differences in acoustic impedance beneath the Earth's surface. Both reflection and refraction seismic techniques are used in groundwater investigations.

Gravity and Magnetic Methods:

Gravity and magnetic methods do not respond directly to water beneath the Earth's crust but can provide valuable information about geological structures that influence groundwater flow and accumulation.

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3.2. REMOTE SENSING TECHNIQUES FOR GROUNDWATER EXPLORATION:

Remote sensing technologies use sensors to measure energy reflected or emitted from the Earth's surface (Sharad, 2021). Various remote sensing techniques are used in groundwater exploration, such as optical remote sensing, thermal remote sensing and radar remote sensing, and play a crucial role in groundwater investigation. Optical remote sensing uses visible, near-infrared, and short-wave infrared wavelengths to analyze surface features related to groundwater occurrence, such as geological structures and vegetation patterns; Thermal infrared remote sensing can detect surface temperature fluctuations that may indicate groundwater drainage zones or shallow water tables, and radar remote sensing can measure surface deformations associated with groundwater extraction or recharge, enabling the assessment of changes in groundwater storage.

A variety of methods can be used to remotely measure groundwater, including ground-based, airborne, satellite-based, and gravity-based methods.

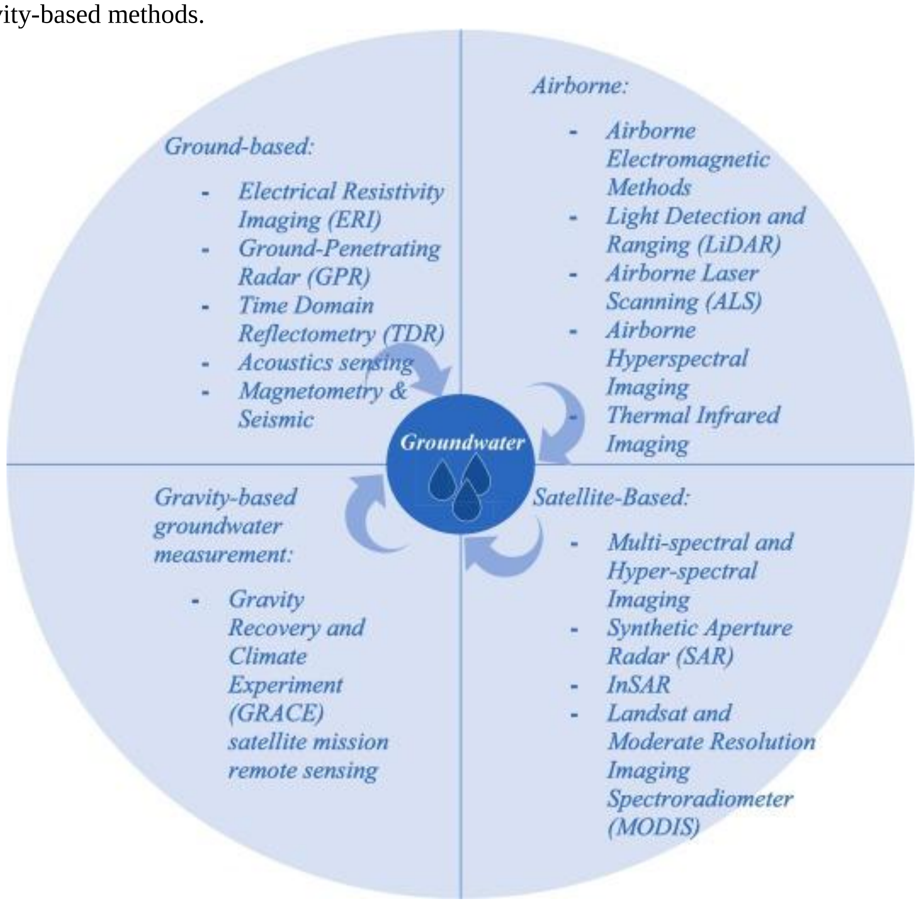

Fig. 2. Summary of remote sensing technologies that can be applied to groundwater studies (Ibrahim *et al.*, 2024)

5.0 CASE STUDIES

Case Study 1: Groundwater prospecting using remote sensing and geoelectrical methods in the North Cameroon (Central Africa) metamorphic formations.

This study by Gouet*et al.*, (2021) demonstrates the integration of remote sensing and geoelectrical techniques for groundwater exploration in the metamorphic formations of North Cameroon.

Methodology:

1. Remote Sensing:

The study used Landsat 8 imagery, applied automatic and manual lineament extraction techniques and generated lineament density maps.

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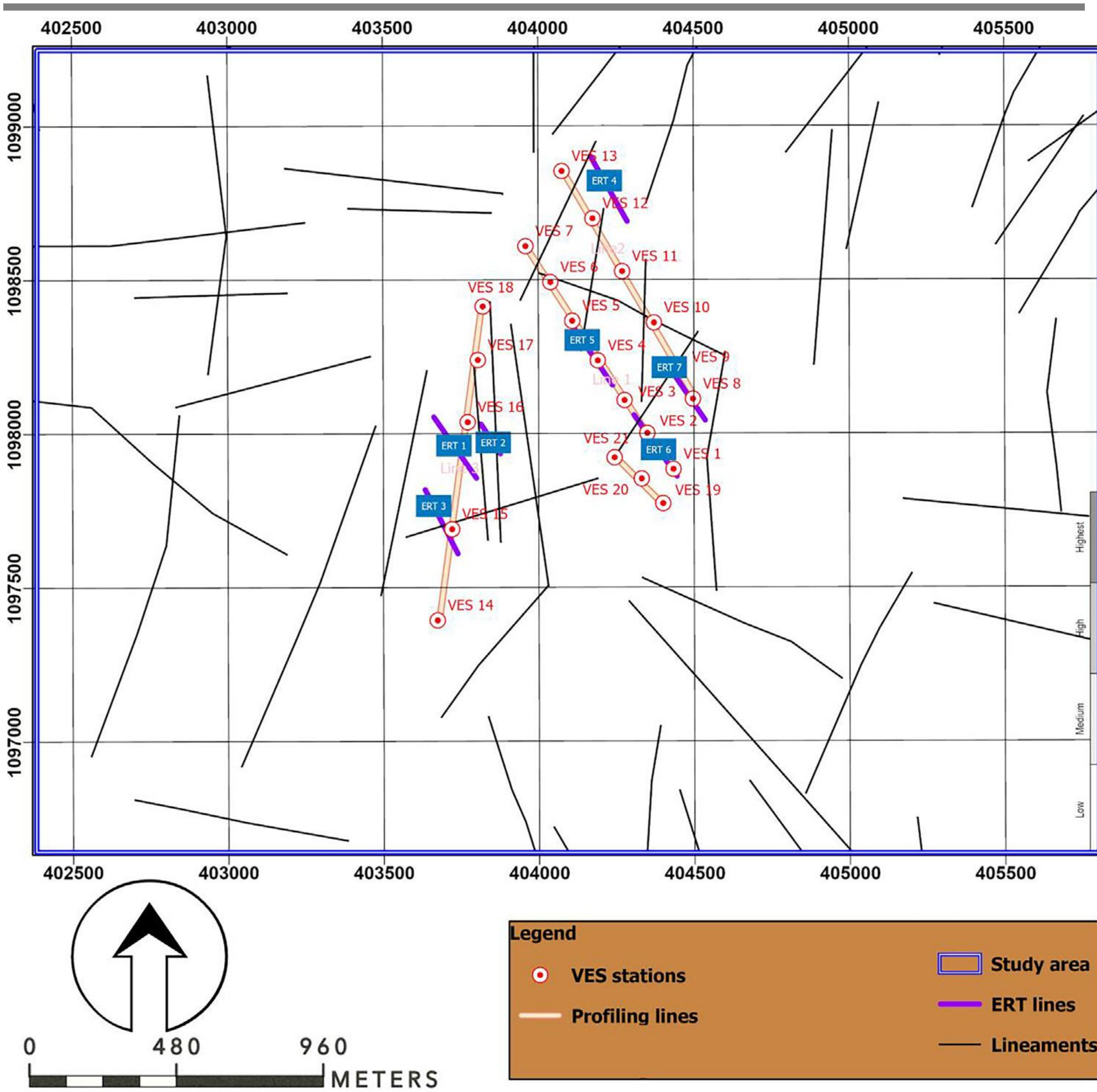

Fig. 3: Lineament map of the study area.

2. Geoelectrical Surveys:

Conducted vertical electrical sounding (VES), electrical resistivity profiling, and electrical resistivity tomography (ERT). Surveys were performed perpendicular to major lineament orientations.

Key Findings:

Lineament Analysis: Extracted 61 lineaments with a total length of 38,853. The main lineament directions, N-S and N22°E (NNE-SSW) were identified. Secondary directions is N10°E, NE-SW, and W-E to N100°E

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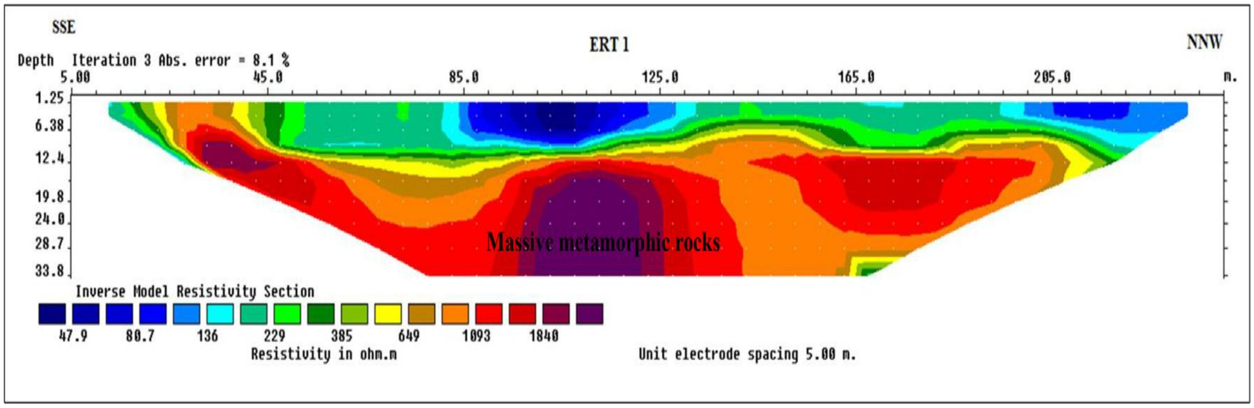

Fig. 4: Electrical resistivity tomography (ERT) along profiles

Lineament Density: High-density areas (626–938 m/km²) are located in the centre and western parts. These areas were correlated with marble and schist formations.

Geoelectrical Results: VES revealed 3-5-layer models of topsoil, metamorphic rocks, and basement. ERT profiles identified conductive discontinuities associated with fractured zones. Low resistivity zones (< 1000 Ω.m) indicated potential groundwater targets.

The integrated approach effectively delineated potential groundwater zones. Fractured marble formations were identified as the most promising aquifer targets. The study demonstrated the complementary nature of remote sensing and geoelectrical methods for groundwater exploration in metamorphic terrains. This case study highlights the successful application of combined remote sensing and geophysical techniques for groundwater prospecting in challenging geological environments.

Case Study 2: Groundwater possibilities in Wadi Ibib Basin using Geophysical Exploration and Remote Sensing Techniques.
This case study examines groundwater exploration efforts in Wadi Ibib, located in the southeastern corner of Egypt between latitudes 22° 29' 00" - 22° 45' 15" N and longitudes 35° 29' 15" - 35° 45' 10" E. It aimed to identify potential groundwater resources and suitable drilling locations along the Wadi's main channel using an integrated approach combining geomorphological analysis, geological mapping, and geophysical surveys.

Methods:
Geomorphological and morphometric analysis using digital elevation models and satellite imagery
Geological mapping and structural analysis
Land magnetic survey along a 35 km profile of the Wadi channel
33 vertical electrical soundings (VES) along the Wadi channel

Key Findings:
- Morphometric analysis indicated favorable conditions for groundwater recharge, including low slope, high stream order, and elongated basin shape.
- Geological mapping revealed three primary aquifer types which are Quaternary alluvial deposits, Nubian Sandstone (Upper Cretaceous), and Fractured basement rocks
- Geophysical surveys identified the variable depth to the basement from 0-300+ m along the wadi. Several sub-basins with depths of 85-160 m are suitable for groundwater accumulation.
- Two main water-bearing layers:
  a) Conglomerate and sandstone layer (40-100 Ohm-m)
  b) Fractured basement (190-480 Ohm-m)
- Seven promising locations were identified for drilling new water wells, with recommended depths of 100-150 m.

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This integrated approach allowed for the comprehensive characterization of the hydrogeological conditions and identification of the most favorable drilling targets along Wadi Ibib. The case demonstrates the value of combining geomorphological, geological, and geophysical methods for groundwater exploration in arid regions (Hamed & Gergis, 2016).

Case Study 3: Geophysical Investigations and Remote Sensing Techniques for Groundwater Exploration in Wadi Almilk Area, North Kordofan State, Sudan

Zeinelabdein and Elsheikh (2015) conducted a study to investigate groundwater occurrence in the Wadi Almilk area of North Kordofan State, Sudan. This semi-arid region faces acute shortages of fresh water supply despite being an essential area for gum Arabic production and livestock. The study employed an integrated methodology combining remote sensing, structural analysis, and geophysical techniques to identify potential groundwater zones in the predominantly basement rock terrain.

Methodology:
Remote sensing analysis of Landsat 8 OLI imagery and SRTM data
Lineament mapping and structural analysis
Electrical resistivity surveys:
  31 resistivity profiles using Wenner configuration
  30 vertical electrical soundings (VES) using Schlumberger array

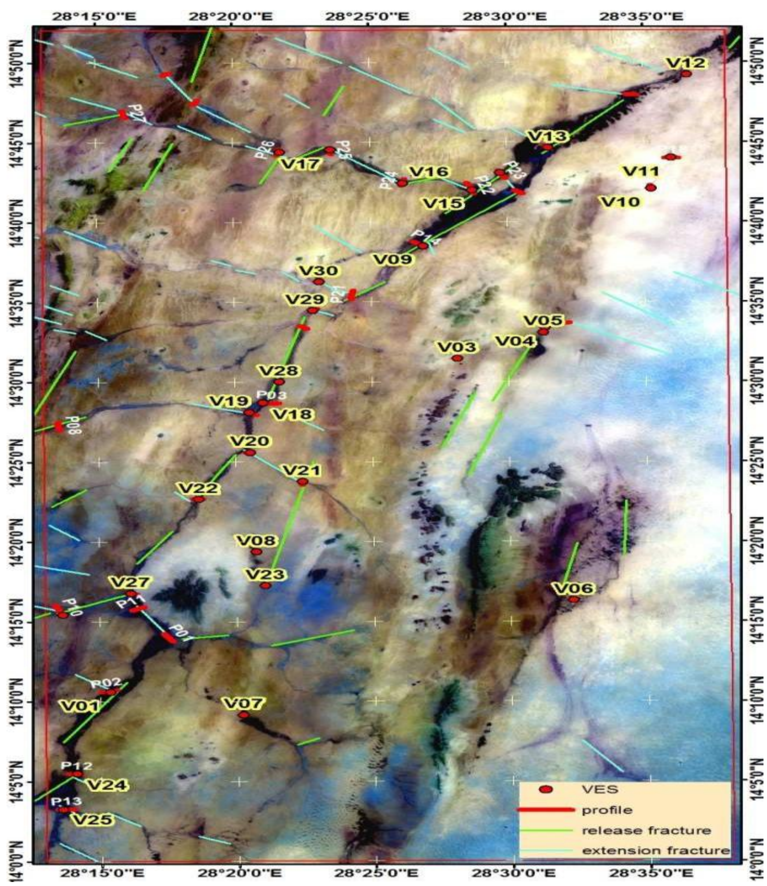

Fig. 5:The geophysical measurement locations.

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Key Findings:

Structural analysis revealed that fractures trending ESE-WNW (115°-135°) and NNE-SSW (15°-35°) were most favorable for groundwater accumulation, especially at their intersections.

Resistivity profiling identified conductive zones with values of 40-100 Ωm, interpreted as weathered or saturated fractured basement rock zones.

VES curves showed different types (Q, H, KH) representing alluvial sediments, weathered basement, fractured basement, and hard basement rocks.

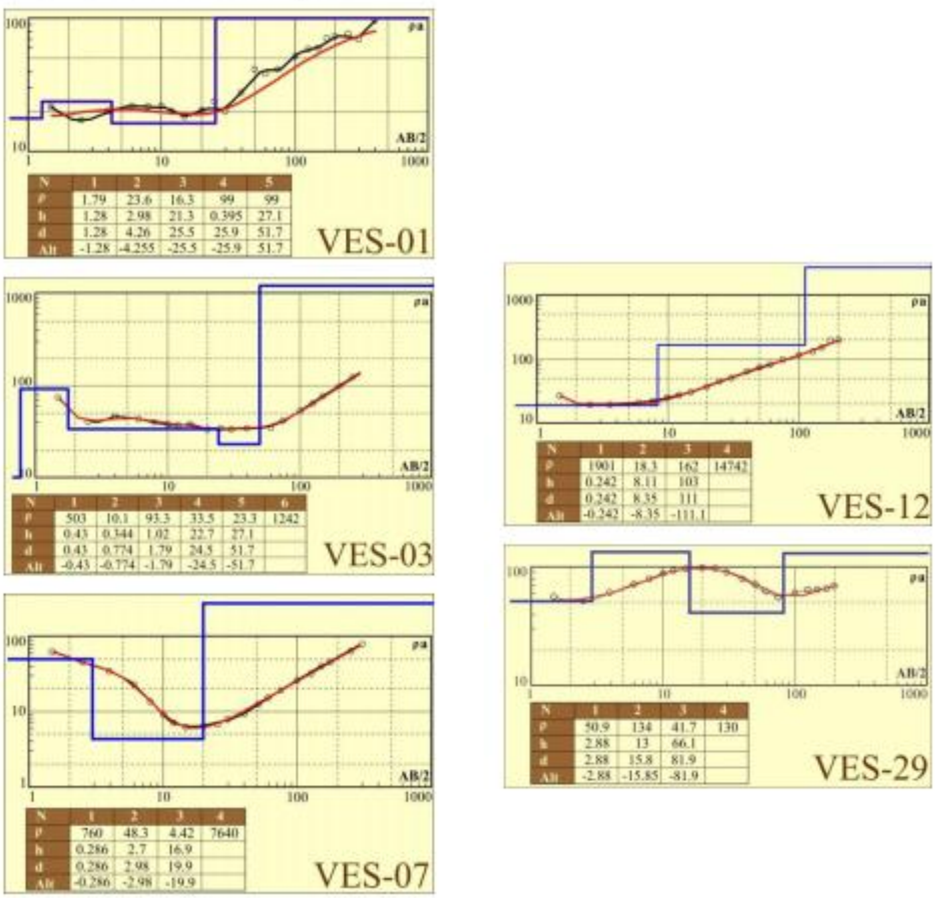

Fig. 6. Curves of selected measured Vertical Electrical Sounding showing the major lithological varieties in the study area.

Based on the integrated analysis, 14 out of 30 measured VES points were identified as potential sites for groundwater occurrence. The study demonstrated the effectiveness of combining remote sensing, structural analysis, and geophysical methods to guide groundwater exploration in basement terrain. This integrated approach allowed for more targeted and efficient identification of promising drilling locations in a challenging hydrogeological environment.

6.0 Challenges and Limitations in Groundwater Exploration and Management Using Geophysical and Remote Sensing Techniques

Despite the significant advances in geophysical and remote sensing techniques for groundwater exploration and management, several challenges and limitations impact their effectiveness. These include issues related to interpretation, depth of investigation, resolution, urban noise and the need for an objective investigation. This

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section discusses these challenges in detail, supported by concrete examples and relevant references from recent literature.

Non-Uniqueness of Geophysical Interpretations

One of the biggest challenges with geophysical methods is the ambiguity of geophysical interpretations. Different subsurface structures and materials can produce similar geophysical signatures, making it difficult to identify specific features. For example, low resistivity anomalies detected by electrical resistivity methods may indicate the presence of groundwater, clay, or saline water, each requiring different management approaches (Loke *et al.*, 2015).
To address this problem, integrating multiple geophysical methods and complementing them with geological and hydrogeological data is crucial. For example, in the Wadi Ibib Basin study, vertical electrical soundings (VES) and magnetic surveys were combined to reduce ambiguity, although soil testing through well drilling was still required to confirm the results (Hamed & Gergis, 2016).

Limited Depth of Investigation for Some Methods

The depth of investigation of various geophysical methods can limit their effectiveness, especially for deep aquifers. Methods such as ground-penetrating radar (GPR) are suitable for shallow-depth surveys but are limited to depths of a few meters to tens of meters, depending on soil and rock conditions (Jol, 2009). Deeper aquifers require methods such as electrical resistivity measurements and seismic surveys, which allow greater depth penetration but come with greater complexity and cost.

In regions with deep groundwater systems, the exclusive use of shallow geophysical methods may produce incomplete or misleading results. For example, the Wadi Ibib Basin study used VES. It provided subsurface information to depths of over 150 meters, which was sufficient for this environment but may not be sufficient in other regions requiring more in-depth investigations (Hamed & Gergis, 2016).

Resolution and Scale Issues in Remote Sensing Data

Remote sensing techniques provide valuable large-scale data for identifying regional geological and hydrological features. However, the resolution of remote sensing data can be a significant limitation, particularly in detecting smaller features that are critical for detailed groundwater studies. While high-resolution satellite images and aerial photography provide detailed surface information, their resolution is often insufficient to characterize the subsurface without complementary geophysical data.

For example, multispectral satellite data such as Landsat are invaluable for mapping large-scale features such as fault lines and lithological boundaries (Gao, 2015). However, these data may not resolve finer details, such as small fractures or localized groundwater seeps, which are critical to accurate groundwater assessments. Higher-resolution remote sensing data from advanced satellites such as Sentinel-2 can partially solve this problem, but it still requires integration with ground-based geophysical and geological data (Chen *et al.*, 2018).

Influence of Cultural Noise in Urban Areas

Cultural noise significantly affects the quality and reliability of geophysical data in urban and densely populated areas. Cultural noise includes electromagnetic interference, vibration, and other disturbances caused by human activities such as traffic, construction, and electrical infrastructure. These noises can obscure or distort geophysical signals, making subsurface features difficult to interpret.

For example, studies of urban groundwater using electrical resistivity or magnetic measurements can be interfered with by buried utilities, reinforced concrete structures, and metal objects, resulting in anomalies unrelated to geological or hydrological features (Milsom & Eriksen, 2015). This interference requires careful data processing and often limits the applicability of certain geophysical methods in urban environments. In the Wadi Ibib basin, cultural noise was less of a problem due to the remote location, allowing clearer geophysical measurements (Hamed & Gergis, 2016).

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Need for Ground-Truthing and Calibration

Geophysical and remote sensing data must be calibrated and validated against ground truth data to ensure accuracy. Ground truthing involves collecting direct observations and measurements in the field, such as by drilling boreholes, sampling water, and conducting physical investigations. This process is essential for correlating geophysical signatures with actual geological and hydrological conditions.

Without solid data, geophysical and remote sensing interpretations remain speculative. For example, in the Wadi Ibib Basin study, the identification of aquifer zones and the assessment of groundwater potential were initially based on geophysical data. However, these interpretations were validated through drilling and water sampling, confirming the presence of groundwater and refining the geophysical models (Hamed & Gergis, 2016).

On-site investigation is resource-intensive and time-consuming and presents significant logistical challenges, particularly in remote or inaccessible areas. Despite these challenges, ensuring the reliability and usability of geophysical and remote sensing data remains critical to any groundwater exploration and management strategy.

7.0 DISCUSSION

Synthesis of Main Findings from the Reviewed Literature

The integration of geophysical and remote sensing techniques has significantly advanced groundwater exploration and management, enabling detailed insights into subsurface properties that are not achievable using traditional methods alone. The reviewed literature highlights the effectiveness of various geophysical methods such as electrical resistivity tomography (ERT), ground penetrating radar (GPR), seismic methods, electromagnetic methods and gravity-based techniques. Each of these methods offers unique advantages in identifying and characterizing groundwater resources. For example, ERT and GPR are particularly effective at delineating subsurface features and detecting water-bearing formations, while seismic methods enable high-resolution imaging of geological structures.

Remote sensing techniques, including optical, thermal and radar-based methods, complement geophysical approaches by providing comprehensive spatial coverage and temporal monitoring capabilities. These techniques are invaluable for mapping surface features, monitoring land use and vegetation changes, and detecting signs of subsurface water at the surface. Integrating remote sensing data with geophysical survey results increases the accuracy and efficiency of groundwater surveys and enables more informed decision-making in water resource management.

Recent advances in data collection, processing, and interpretation have further improved the reliability and applicability of these techniques. Innovations such as improved inversion algorithms for ERT, improved signal processing for GPR, and the development of airborne geophysical systems have expanded the scope and depth of groundwater exploration. Additionally, using machine learning and other advanced computational methods to integrate and interpret diverse data sets has shown promising results in reducing ambiguity and improving model accuracy.

Agreements and Contradictions in the Field

The reviewed literature generally agrees on the effectiveness of combining geophysical and remote sensing techniques for groundwater exploration. Integrating these methods allows for a more comprehensive understanding of subsurface conditions and enables precise identification of potential groundwater zones. Many studies emphasize the synergistic benefits of using multiple techniques and suggest that this approach leverages the strengths of each method to overcome the limitations of individual techniques.

However, there are also contradictions and areas of debate within the field. A major point of contention is the variability in the effectiveness of certain techniques in different geological environments. While some methods, such as ERT and GPR, are widely used and effective in various settings, others may be more limited in scope. For example, the applicability of seismic methods can be limited by factors such as subsurface heterogeneity and the presence of noise in data acquisition.

Another area of disagreement concerns the interpretation of geophysical data, particularly in complex hydrogeological environments. Integrating multiple data sets can sometimes produce conflicting results and requires careful analysis and cross-validation to ensure accuracy. Furthermore, the high cost and technical expertise required for some advanced geophysical and remote sensing techniques can be prohibitive, limiting their accessibility and widespread adoption, particularly in resource-limited regions.

Implications of the Findings

The results of this review have significant implications for groundwater exploration and management. The proven effectiveness of integrated geophysical and remote sensing techniques highlights the need to use these methods more widely in groundwater investigations. By providing detailed and accurate subsurface information, these techniques can help identify and characterize groundwater resources more efficiently, leading to better-informed water management practices.

One of the most important implications is the potential for sustainable groundwater management. The ability to accurately map and monitor groundwater resources can support the development of strategies to manage these resources more effectively, thereby ensuring their long-term sustainability. For example, identifying recharge zones and monitoring groundwater levels can be the basis for policies and practices to balance groundwater abstraction and recharge.

Furthermore, integrating these techniques can facilitate the identification of new groundwater resources in water-stressed regions. This is particularly important in arid and semi-arid regions with limited surface water resources and high groundwater demand. Applying geophysical and remote sensing methods in such regions can help locate previously unknown aquifers and provide important water supplies for drinking water, agriculture and industry.

The review also highlights the importance of continued technological advances and innovations in the field. Continuous development of new methods and refinement of existing techniques are critical to improving the accuracy and efficiency of groundwater exploration. Advanced computational methods, such as machine learning for data integration and interpretation, promise to further improve groundwater investigation reliability.

8.0 FUTURE DIRECTIONS

Identifying Gaps in Current Knowledge

Despite significant progress in integrating geophysical and remote sensing techniques for groundwater exploration and management, several gaps remain in current knowledge. Addressing these gaps is critical to further improving the effectiveness and accuracy of these methods.

1. Spatial and Temporal Resolution: One of the main limitations is the spatial and temporal resolution of both geophysical and remote sensing data. While high-resolution methods such as electrical resistivity tomography (ERT) and ground penetrating radar (GPR) provide detailed information about the subsurface, their application is often limited to small areas. In contrast, remote sensing techniques provide extensive spatial coverage but may not have the fine resolution required for detailed subsurface characterization. Bridging this gap through improved resolution and integration of these datasets remains a key challenge (Parsekian *et al.*, 2015).

2. Data Integration and Interpretation: Integrating different geophysical and remote sensing datasets presents significant challenges. Different methods often produce different data types and require sophisticated algorithms and models for effective integration and interpretation. While recent advances in machine learning and data fusion techniques are promising, further research is needed to develop robust, user-friendly tools that can seamlessly integrate and interpret data sets from multiple sources (Karimpouli*et al.*, 2018).

3. Hydrogeological Variability: The effectiveness of geophysical and remote sensing techniques can vary significantly depending on the hydrogeological environment. More systematic studies are needed to evaluate the performance of these methods in different geological contexts, particularly in complex environments such as fractured rock aquifers and heterogeneous alluvial systems. Understanding these variabilities can inform the development of tailored approaches for different environments (Bermejo *et al.*, 2018).

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4. Cost and Accessibility: The high cost and technical expertise required for advanced geophysical and remote sensing techniques can be prohibitive, particularly in developing regions. Research into cost-effective and accessible methods is essential to ensure that these technologies can be widely adopted. This includes the development of low-cost sensors, simplified data processing tools and training programs for local practitioners (Auken *et al.,* 2017).

5. **Long-Term Monitoring:**The sustainability of groundwater resources depends on continuous monitoring. However, long-term monitoring programs using geophysical and remote sensing techniques are still limited. The development of standardized protocols and affordable technologies for long-term groundwater monitoring is critical for assessing the impacts of climate change, land use change, and human activities on groundwater resources (Podgorski *et al.,* 2020).

Suggesting Areas for Future Research

To address these gaps and improve the application of geophysical and remote sensing techniques in groundwater exploration and management, several areas for future research are suggested:

1. Improved data fusion techniques: Future research should focus on developing advanced data fusion techniques that can effectively integrate high-resolution geophysical data with long-range remote sensing data. This includes using machine learning and artificial intelligence to automate data processing and improve the accuracy of subsurface models (Camporese*et al.,* 2015).

2. Multi-Scale Approaches: Research should explore multi-scale approaches that combine the strengths of different methodologies. For example, integrating high-resolution ground-based geophysical surveys with satellite-based remote sensing can lead to a comprehensive understanding of groundwater systems at local and regional scales. Such approaches can increase the accuracy of groundwater potential maps and improve resource management strategies (Costall *et al.,* 2020).

3. Field Validation and Case Studies: Further field validation studies and case studies are needed to evaluate the performance of integrated geophysical and remote sensing techniques in different hydrogeological environments. These studies are intended to document successful applications, identify challenges and provide best practice guidelines. This can help develop tailored approaches for different environments and improve the overall reliability of these methods (Flores Orozco *et al.,* 2019).

4. Development of Cost-Effective Technologies: Research should prioritize the development of cost-effective technologies and methods that can be easily adopted in resource-limited regions. This includes the development of cost-effective sensors, mobile data collection systems and user-friendly data analysis software. Ensuring that these technologies are accessible and affordable is key to their widespread adoption (Mukherjee *et al.,* 2020).

5. Long-Term Monitoring and Sustainability Studies: Long-term monitoring studies using integrated geophysical and remote sensing techniques are essential for understanding the dynamics of groundwater systems over time. Future research should focus on developing protocols and technologies for continuous monitoring, assessing environmental change impacts, and assessing groundwater resources' sustainability. This can provide valuable data for informed decision-making and policy development (Kennedy *et al.,* 2016).

6. Interdisciplinary Collaboration: Promoting interdisciplinary collaboration between geophysicists, hydrologists, remote sensing experts and data scientists is crucial to progress in this field. Collaborative research initiatives can lead to the development of innovative solutions and comprehensive approaches to groundwater exploration and management. Promoting knowledge sharing and collaboration across disciplines can drive progress and promote new insights (Schmutz *et al.,* 2019).

9. CONCLUSION

Geophysical and remote sensing techniques provide powerful tools for groundwater exploration and management, providing valuable data over large areas and at different depths. However, these methods face several challenges,

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including the ambiguity of geophysical interpretations, the limited depth of investigation, resolution and scaling issues in remote sensing data, the influence of cultural noise in urban areas, and the need for ground truth and calibration. Addressing these challenges requires an integrated approach that combines multiple geophysical methods, remote sensing data, and comprehensive ground investigations to improve the accuracy and reliability of groundwater assessments.

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